# App Development Project Report

### 1 Student Details

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**About Me:** I am pursuing dual degree one from IIT Madras (BS in Data Science) and second from Chandigarh University (BE in Cse specialization in AIML). I have web development skills and i am now having experience of building app of more than 6 months.

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### 2 Project Details

**Project Title:** Trekking Management Application

**Problem Statement:** Adventure organizations require efficient systems to manage trekking activities involving trek organizers, staff, and participants. Currently, many trekking groups rely on spreadsheets, phone calls, or manual coordination, which makes it difficult to manage trek approvals, track bookings, avoid overbooking, and maintain trek history.
You are required to build a Trekking Management Application web application that allows Admin, Trek Staff, and Users (Trekkers) to interact with the system based on their roles.

**Approach:** I built the app using flask framework using mvc model and to make the app interactive used css and added effects like hover on objects using css.

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### 3 AI/LLM Declaration

I didn’t used and AI/LLM for this project. Firstly i learned the concepts from w3school and applied those concepts. Sometime i failed difficulties then i took help from documentation and debug the issue for the problem and solved them at own.

No use of AI/LLM

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## 4 Technologies and Frameworks Used

## Technology / Library

## Flask

## SQLAlchemy

## Jinja2

## Html

## Css

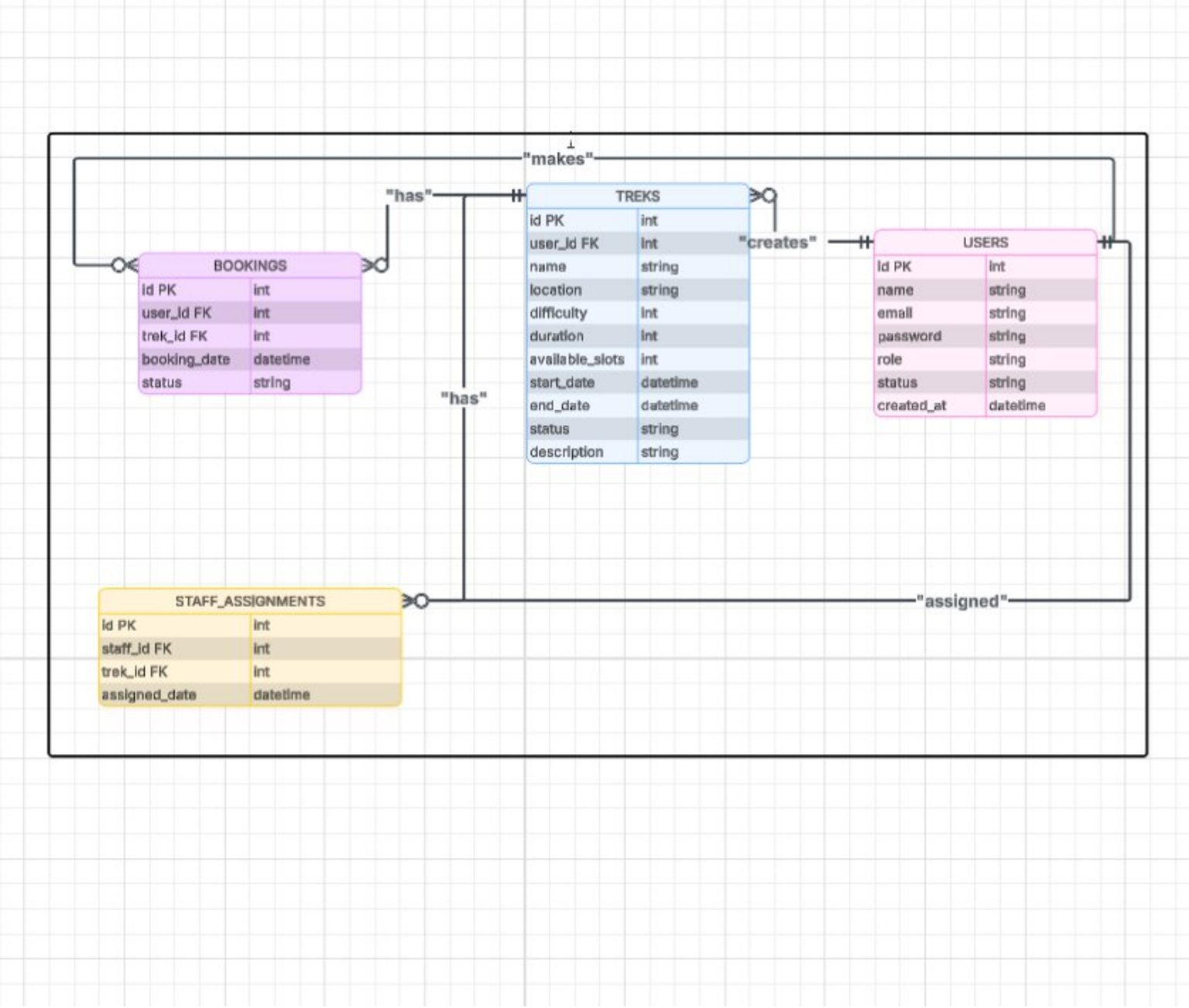
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## 5 Database Schema

### Tables

1. **users**
2. **treks**
3. **bookings**
4. **staff_assignment**

### Er Diagram



**6 API Resource Endpoints**

| Endpoint | Method | Description |
|---|---|---|
| /login | POST | Authenticate user and redirect to the appropriate dashboard |
| /register | POST | Register a new user or staff account |
| /admin | GET | Access the administrator dashboard |
| /staff | GET | Access the staff dashboard |
| /user | GET | Access the user dashboard |
| /admin/newTrek | POST | Create a new trek and assign staff |
| /admin/trekEdit/<int:trekId> | POST | Update trek information |
| /admin/trekDelete/<int:trekId> | GET | Delete a trek |
| /staff/update/<int:trek_id> | POST | Update trek status and available slots |
| /staff/participants/<int:trek_id> | GET | View participants of an assigned trek |
| /user/book/<int:trek_id> | GET | Book a trek |
| /user/cancelBook/<int:trek_id> | GET | Cancel a trek booking |
| /user/edit | GET, POST | Edit user profile |

**7 Video Presentation**

**Drive Link:** Video Link